

Cross-column RT calibration report

METHOD	cross-column calibration onto a reference column (v2)
REFERENCE COLUMN	Column 25 (bundled default)
STANDARDS PANEL	mix15
STAGE 1 CURVE	595 m/z-matched pairs (173 standards-run + 422 sample-run), 433 kept after outlier trimming
SERIES TERM	series term gated off: leave-own-out MSE 35% above curve-only (threshold: 20% below)
STAGE 2 ANCHORS	not requested (use_sample_anchors=False) - stage 2 is opt-in since 1.2.2
IRT SCALE	1-100, affine on the reference RT axis between the earliest and latest of 14 landmarks (1.46-21.61 min); landmark panel mix15

FEATURES

samples	22,314
standards run	8,768
feature rows in each table	

MATCHED PAIRS

standards	173
sample	422
433 of 595 kept after MAD trimming	

CURVE RESIDUAL

median	0.389
P90	4.245
reference – fitted (min)	

SERIES TERM

off

35% worse than the curve alone · stage-1 curve only

IRT LANDMARKS

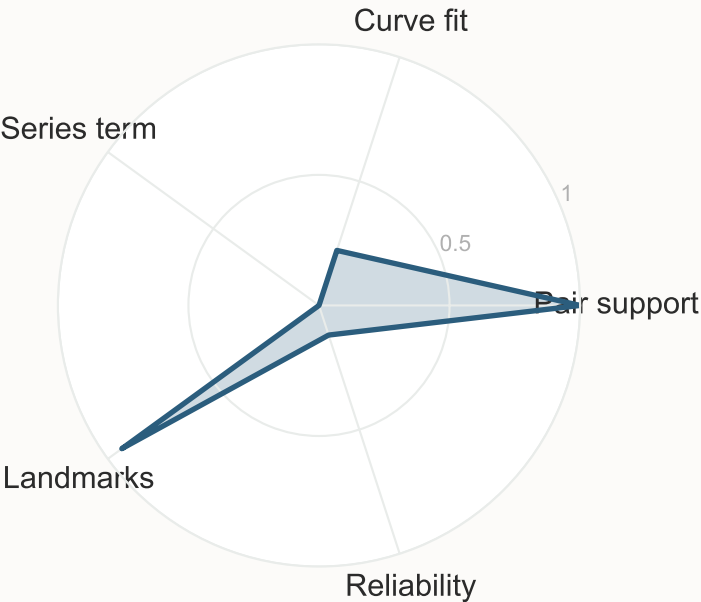
14

RT 1.46-21.61 min on the reference run

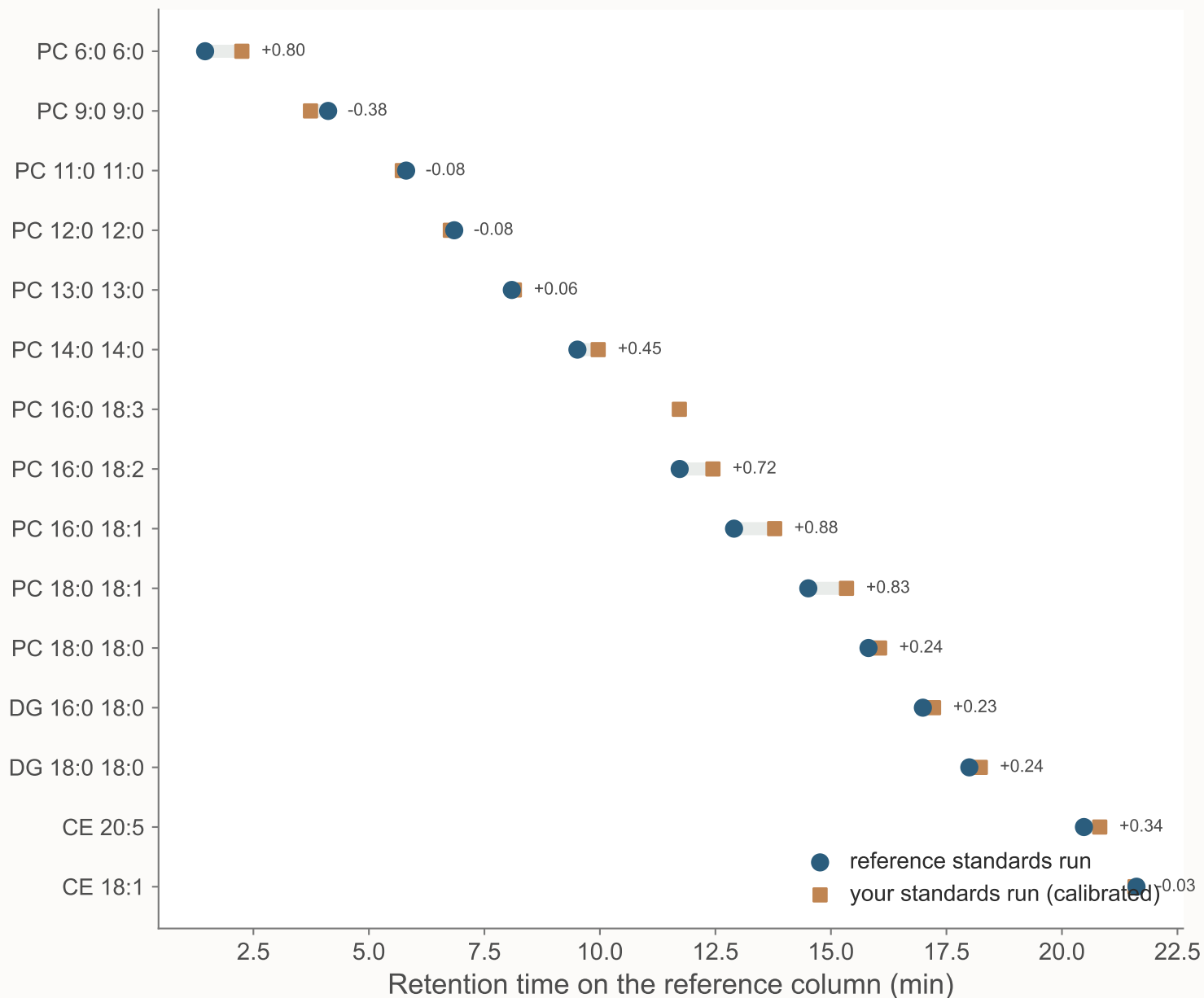
OUTPUT

iRT range	-6-100
extrapolated	2.5%
12% high-reliability	

Quality fingerprint

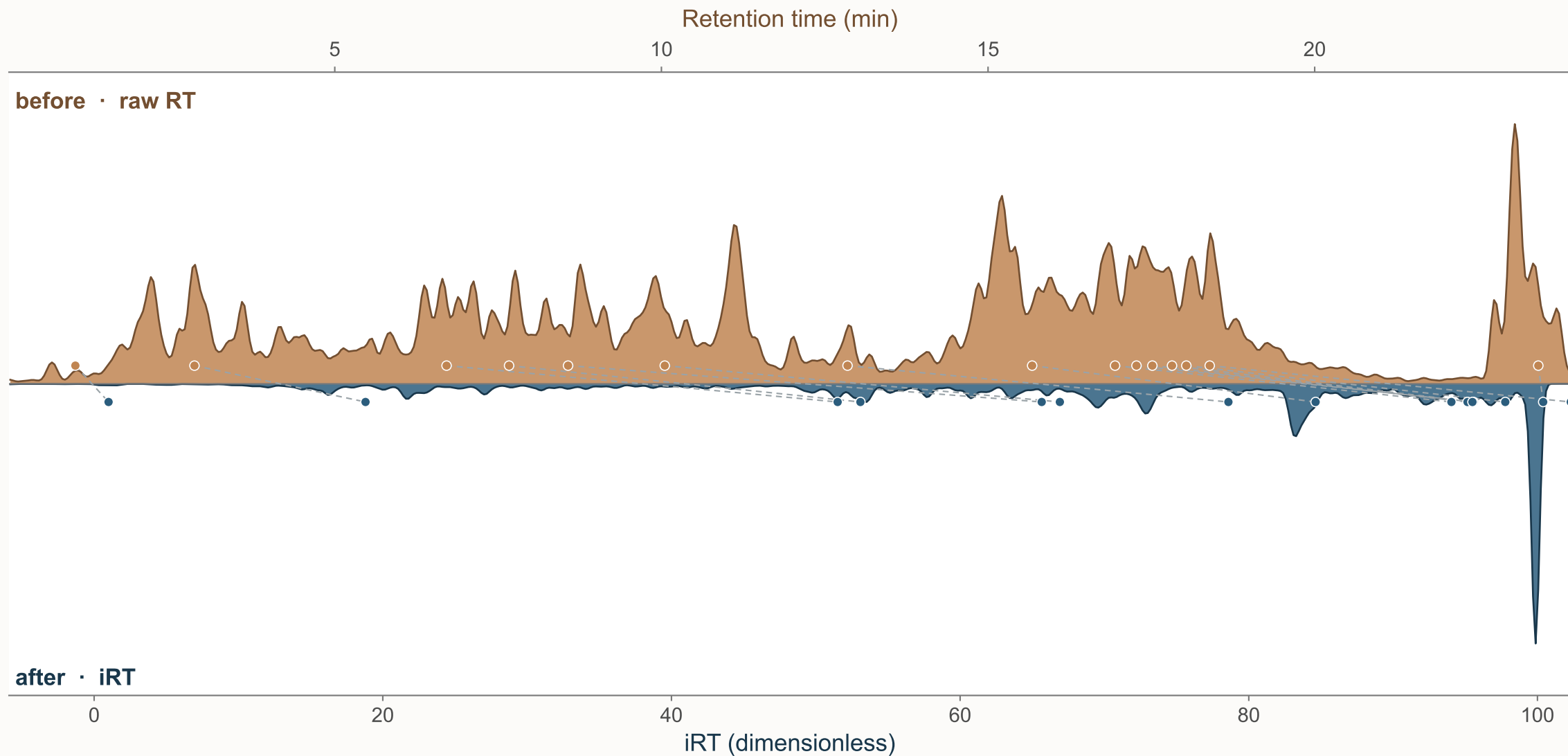


Standard detection — reference vs panel vs samples



Reconstructed feature-intensity profile · before vs after calibration

sampled matched pairs connected — line slant = RT offset · reconstructed from the feature table (binned intensity)



Cross-column calibration curve · your column -> the reference column

series term: series term gated off: leave-own-out MSE 35% above curve-only (threshold: 20% below)

